PERSONA BASED INFORMATION RETRIEVAL

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STRUCTURE

We used the **antique/test** dataset.
The original 200 queries and 403,666 documents resulted in excessively long execution times, so we decided to reduce the dataset.

INTRODUCTION

Data loading and cleaning

- Loading ***queries***, ***docs***, ***qrels*** dataframes

Explorative Data Analysis

- Query response distribution
- Distribution of ***response*** and ***queries*** lengths (in words)
- Top word frequency in ***responses*** and ***queries*** (excluding stop words)

PHASE I

- Baseline retrieval (BM25)
- Neural reranking
- Functions section: evaluation metrics – precision, recall, MAP, nDCG
- Evaluation metrics
  - BM25: dataset preparation, precision and recall, MAP, nDCG
  - Neural reranking: dataset preparation, precision and recall, MAP, nDCG

PHASE II

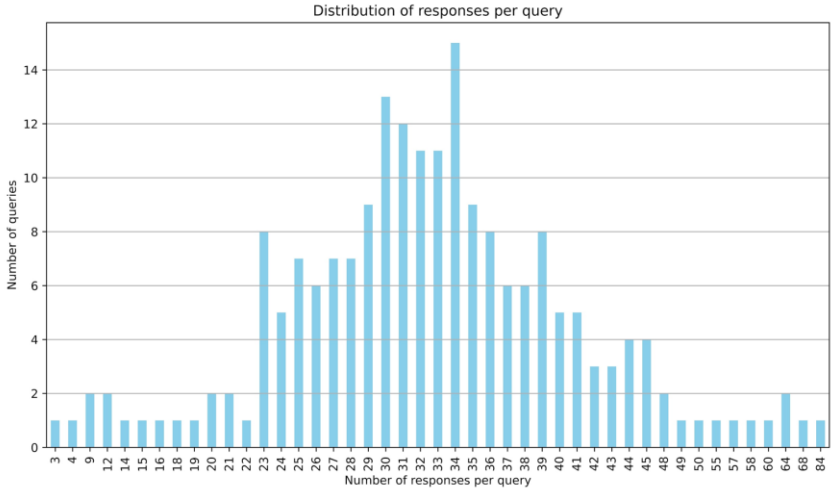
Query expansion, novice, expert

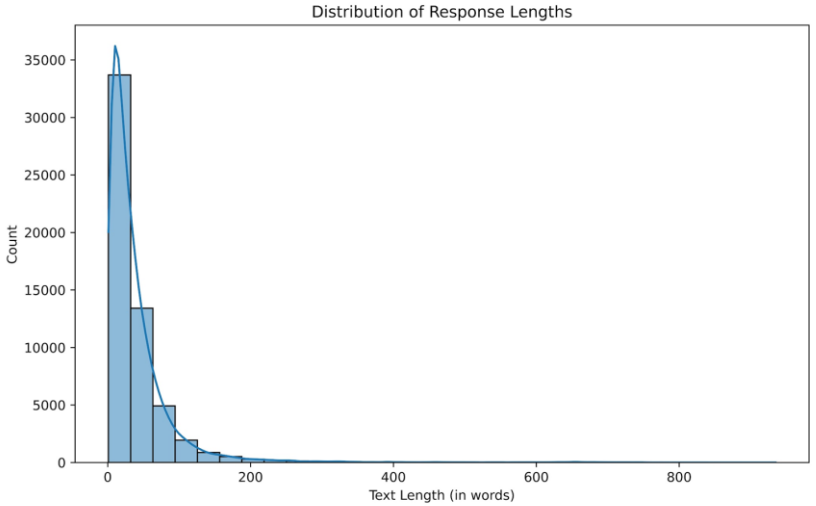
- BM25
- Neural reranking
- Evaluation metrics
  - BM25: dataset preparation, precision and recall, MAP, nDCG
  - Neural reranking: dataset preparation, precision and recall, MAP, nDCG

EXPLORATIVE DATA ANALYSIS - 1

The EDA is focus on:

- Query response distribution in *df_qrels*
- Distribution of response lengths (in words)

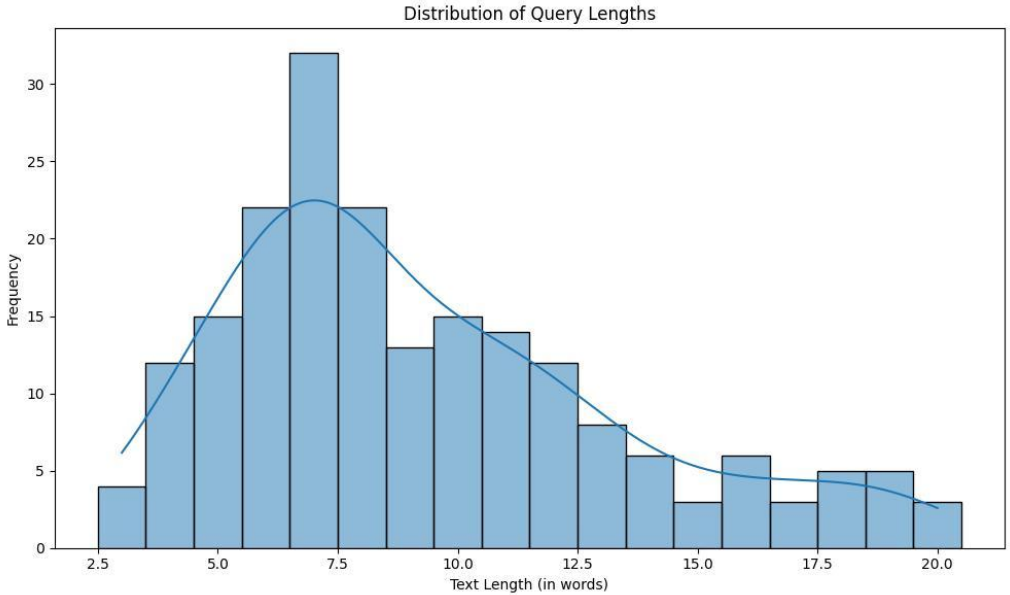




EXPLORATIVE DATA ANALYSIS - 2

The EDA is focus on:

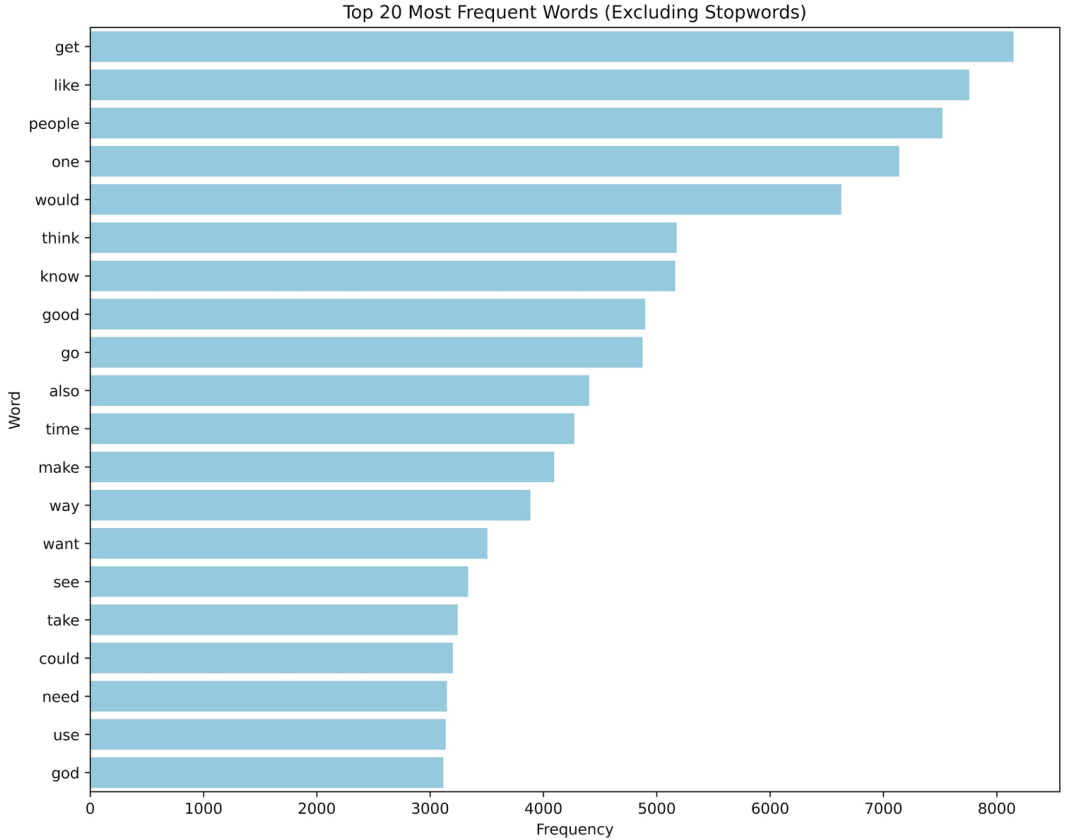
- Distribution of query lengths (in words)



EXPLORATIVE DATA ANALYSIS - 3

The EDA is focus on:

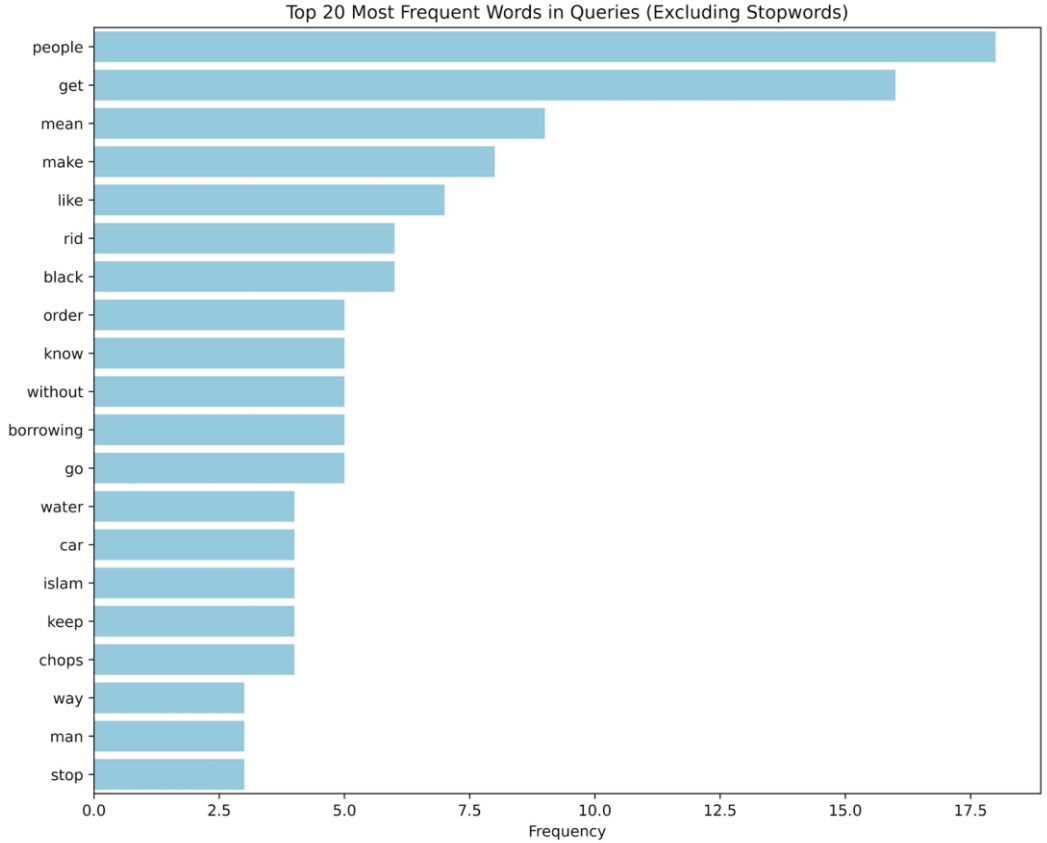
- Top word frequency in responses (excluding stop words)



EXPLORATIVE DATA ANALYSIS - 4

The EDA is focus on:

- Top word frequency in queries (excluding stop words)



TYPES OF DATASET

For the automatic generation of the *expansion*, *novice* and *expert* queries, the Gemini 2.0 Flash Lite model was used, a free language model developed by Google.

query_id	text	query_expanded	query_novice	query_expert
3990512	how can we get concentration onsomething?	How to improve focus/attention? Techniques & strategies for concentrating on a task?	How do I focus on stuff?	How to enhance cognitive focus and sustained attention on [topic]?
714612	Why doesn't the water fall off earth if it's round?	Why is water retained on Earth's spherical shape, considering gravity's pull?	How come water stays on Earth, even though it's a ball?	Considering Earth's spheroidal geometry, explain water's adherence, referencing gravity's influence.
2528767	How do I determine the charge of the iron ion in FeCl3?	Determine iron ion charge in ferric chloride (FeCl3) using chlorine's -1 charge.	How do I figure out the iron thing's charge in that FeCl3 stuff?	Determine iron's oxidation state in FeCl3.
821387	I have mice.How do I get rid of them humanely?	"Eradicate mice from my home ethically. Suggest humane removal methods."	How do I get rid of mice? I don't want to hurt them! What should I do?	Implement humane rodent control strategies.

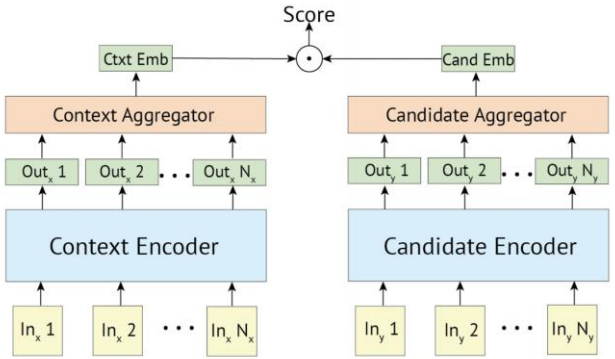
BM25 and NEURAL RERANKING

Best Matching 25 (BM25)

BM25 is a ranking algorithm used to measure how relevant a document is to a given query.
It belongs to the probabilistic model family and considers term frequency and document length.
Common terms are penalized to avoid bias toward frequent but uninformative words.
Its balance of simplicity and effectiveness makes it a strong baseline in search systems.

NEURAL RERANKING (Bi-Encoder)

Bi-Encoder neural reranking models encode the query and document separately into vectors.
Similarity is measured (e.g., via cosine similarity) to assess relevance between the two.
Since document embeddings are independent of the query, they can be precomputed and indexed.
This approach enables fast and scalable retrieval, ideal for large-scale search systems.



(a) Bi-encoder

EVALUATION METRICS

PRECISION
Measures the proportion of retrieved documents that are actually relevant. It evaluates how accurate the system is in returning only relevant results.

RECALL
Measures the proportion of relevant documents that have been successfully retrieved. It reflects the system's ability to find all relevant results.

Mean Average Precision (MAP)
Represents the average of the precision values at the ranks where relevant documents are found, averaged across all queries. It considers both relevance and ranking.

Normalized Discounted Cumulative Gain (nDCG)
Assesses the quality of the ranking by giving higher importance to relevant documents appearing at the top. It is normalized to allow fair comparison across different queries.

RESULTS

BM25

	PRECISION	RECALL	MAP	nDCG
original_dataset	0.24	0.11	0.07	0.47
expanded_dataset	0.15	0.08	0.06	0.23
novice_dataset	0.11	0.05	0.04	0.19
expert_dataset	0.10	0.06	0.04	0.17

NEURAL RERANKING

	PRECISION	RECALL	MAP	nDCG
original_dataset	0.43	0.21	0.18	0.51
expanded_dataset	0.37	0.19	0.16	0.44
novice_dataset	0.26	0.13	0.10	0.32
expert_dataset	0.30	0.16	0.13	0.35